EXPNO: 2

DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS.

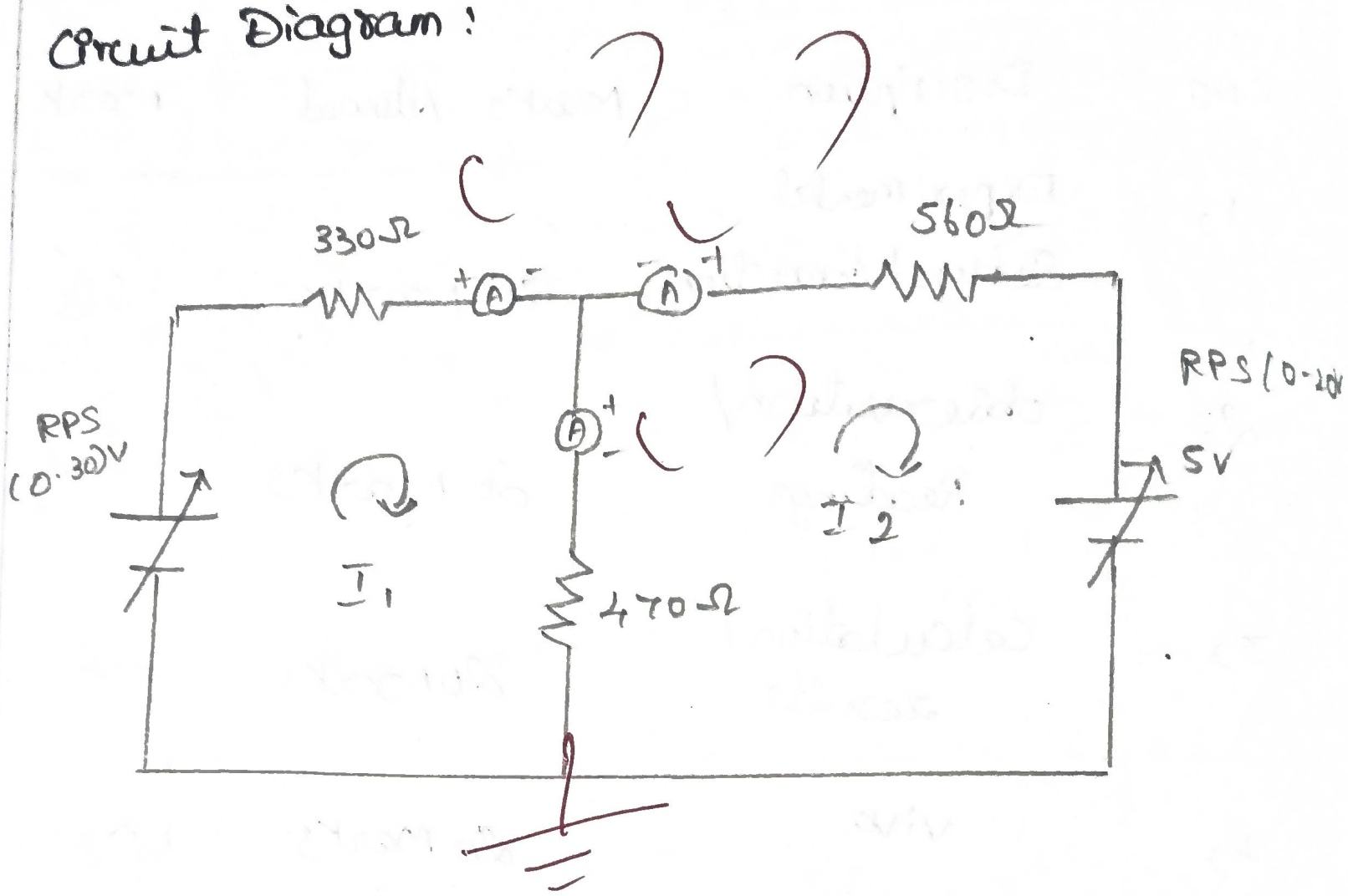
AIM:

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

Apparatus Required:-

SL-NO	Apparatus	Specification	Quantity
1.	Regulated power supply (RPS)	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistor	330Ω, 470Ω 560Ω	each 1
4.	Bread board	–	1

Circuit Diagram :



Tabular Column:

parameter	Theorectical	Practial
I_1	1.2 mA	0.5 mA
I_2	4.2 mA	4.8 mA
$I_1 - I_2$	5.4 mA	5.4 mA

Op verified
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Theoretical Calculation

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} = 330\Omega + 470\Omega = 800\Omega$

$R_{12} = 470\Omega$, $R_{21} = 470\Omega$

$R_{22} = 560 + 470 = 1030\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = 800 \times 1030 - (470)^2$

$= 603100$

$$\Delta_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix}$$

$= 3090 - 2350 = 740$

$$\Delta_2 = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = -4000 + 1410 = -2590$$

$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 1.2\ mA$

$I_2 = \Delta_2 / \Delta = -2590 / 603100 = -4.2\ mA$

$I_1 - I_2 = 1.2 - (-4.2) \Rightarrow 5.4\ mA$

PROCEDURE:

1) Give Connections as per the circuit diagram.

2) Switch on the supply, vary the RPS and set a particular input voltage.

3) Note down the readings of ammeters and voltmeters and tabulate them.

4) Vary the RPS for different input voltage and note down the readings of all the meters

5) Reduce the RPS to its minimum value and switch off the supply

6) using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

Marks split-up:

S.NO	Description	Marks allotted	Mark obtained
1,	Experimental/ Setup/ connection	20 Marks	20
2,	observation / Readings	20 Marks	20
3,	Calculations/ Results	20 Marks	20
4,	viva	20 Marks	20
5,	Record	20 Marks	20
		Total :	100

100

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RESULT:

Thus, the mesh analysis is verified practically and theoretically. The resultant currents for 3 / 5 V supply are:

a) The current across 330Ω resistor is 0.5 mA.

b) The current across 560Ω resistor is 4.8 mA.

c) The current across 470Ω resistor is 5.4 mA.